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METRIC UNITS

METHODS OF
TESTING CONCRETE

Part 4. Methods of testing
concrete for strength

ORIGINAL

BRITISH STANDARDS INSTITUTION

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BS 1881 : Part 4 : 1970

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The Institution desires to call attention to the fact that this British Standard does not purport to include all the necessary provisions of a contract.

In order to keep abreast of progress in the industries concerned, British Standards are subject to periodical review. Suggestions for improvements will be recorded and in due course brought to the notice of the committees charged with the revision of the standards to which they refer.

A complete list of British Standards, numbering over 5000, fully indexed and with a note of the contents of each, will be found in the British Standards Yearbook, price 20s. The BS Yearbook may be consulted in many public libraries and similar institutions.

This standard makes reference to the following British Standards:

- | | |
|----------|--|
| BS 308. | Engineering drawing practice |
| BS 427. | Method for Vickers hardness test
Part 1, Testing of metals. |
| BS 812. | Methods for the sampling and testing of mineral aggregates, sands and fillers. |
| BS 860. | Tables for comparison of hardness scales. |
| BS 882. | Coarse and fine aggregates from natural sources. |
| BS 891. | Method for Rockwell hardness test.
Part 1, Testing of metals. |
| BS 1134. | Centre-line-average height method for the assessment of surface texture. |
| BS 1610. | Methods for the load verification of testing machines. |
| BS | Methods of testing mortars and specification for mortar testing sand (in course of preparation). |

British Standards are revised, when necessary, by the issue either of amendment slips or of revised editions. It is important that users of British Standards should ascertain that they are in possession of the latest amendments or editions.

The following BSI references relate to the work on this standard:
Committee references CEB/4 and CEB/4/3
Draft for comment D65/10400

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CO-OPERATING ORGANIZATIONS

The Cement, Lime and Gypsum Products Industry Standards Committee, under whose supervision this standard was prepared, consists of representatives from the following Government departments and scientific and industrial organizations:

- Association of Consulting Engineers
- British Granite and Whinstone Federation
- *British Precast Concrete Federation
- British Railways Board
- *British Ready Mixed Concrete Association
- British Slag Federation
- British Steel Industry
- *Cement and Concrete Association
- *Cement Makers' Federation
- Chalk Quarrying Association
- Chemical Industries Association
- *Concrete Society Ltd.
- *Federation of Civil Engineering Contractors
- *Greater London Council
- Gypsum Plasterboard Development Association
- *Institution of Civil Engineers
- *Institution of Municipal Engineers
- *Institution of Structural Engineers
- Institution of Water Engineers
- *Limestone Federation
- *London Transport Board
- Ministry of Housing and Local Government
- *Ministry of Public Building and Works
- *Ministry of Public Building and Works—Building Research Station
- *Ministry of Transport
- National Federation of Building Trades Employers
- *Royal Institute of British Architects
- Royal Institution of Chartered Surveyors
- *Sand and Gravel Association of Great Britain
- Society of Chemical Industry
- Welwyn Hall Research Association

The Government departments and scientific and industrial organizations marked with an asterisk in the above list, together with the following, were directly represented on the committees entrusted with the preparation of this standard:

- District Surveyors' Association
- Incorporated Association of Architects and Surveyors
- Ministry of Transport—Road Research Laboratory
- Reinforcement Manufacturers' Association

BRITISH STANDARD METHODS OF TESTING CONCRETE

Part 4. Methods of testing concrete for strength

FOREWORD

The purpose of this British Standard is to provide suitable methods for testing concrete, both on site and in the laboratory, so that individual specifications can include references to this standard instead of giving test methods as appendices. The term 'site' is used when samples are being taken from concrete to be used in the works and the term 'laboratory', when most of the concrete is to be used in the samples. Thus it is possible to provide uniformity in the methods of making and recording the tests carried out on the concrete.

The tests do not apply to concrete whose nominal maximum aggregate size exceeds 40 mm.

The standard has been revised under the authority of the Cement, Lime and Gypsum Products Industry Standards Committee in the light of experience gained in carrying out the tests described in the first edition published in 1952 and in order to expand its scope to cover additional methods of testing concrete.

The standard has been rearranged and is now published in five Parts, issued separately as follows:

- Part 1. Methods of sampling fresh concrete.*
- Part 2. Methods of testing fresh concrete.*
- Part 3. Methods of making and curing test specimens.*
- Part 4. Methods of testing concrete for strength.*
- Part 5. Methods of testing hardened concrete for other than strength.*

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The present Part 4 includes clauses on the test for compressive strength of test cubes, the preparation and compression testing of drilled cores from concrete, the preparation of sawn beams, the test for flexural strength, the test for compressive strength using portions of beams broken in flexure ('equivalent cube' method) and the test for indirect tensile strength of test cylinders.

Clause 2, describing the test for compressive strength of test cubes, has been modified in respect of the use of auxiliary platens where the machine platens are larger than the specimen. A minimum surface hardness of both machine and auxiliary platens is specified. In order to improve the reproducibility of results between laboratories advice is given against the use of special high bearing pressure lubricants.

Definitions of terms describing the compaction of concrete in a core are now given in Clause 3, which specifies methods of preparation and compression testing of cores drilled from concrete.

Clause 4 is an addition to the standard and covers the preparation of sawn beams as test specimens.

The new Clause 5, describing the test for flexural strength, has been expanded to cover sawn beam specimens in addition to those of moulded concrete.

Minor amendments have been made to Clause 6, describing the test for compressive strength using portions of beams broken in flexure ('equivalent cube' method). These amendments affect the thickness and surface hardness of the bearing plates.

Clause 7 describes the procedure for determining the indirect tensile strength of test cylinders by the cylinder splitting method.

There are some methods of test which were considered by the committee and omitted because it was felt that at present there were more than ordinary difficulties in standardizing procedures for carrying them out. One of these tests, which it was considered desirable to include as soon as possible, was an accelerated strength test. The problem in connection with accelerated strength tests is to determine a regime of test which will enable the normal strength at later ages to be forecast with reasonable accuracy for any material. Co-operative trials to investigate this are already being carried out by a number of laboratories. The committee will therefore consider this test again when the co-operative trials are complete and it is hoped that it will be possible to standardize the test and issue it later in the form of an addendum to this standard.

Information on any other methods of test that are thought to be suitable for extensive use, or suggestions for improvement on the specified methods, are welcomed and will be considered in the course of future revision of this standard.

In view of the change to metric in industry this standard is expressed in metric terms.

Apparatus for tests which have not appeared previously in the standard are given in terms of rounded metric values which generally do not relate to any inch sizes.

Values of stress in this standard are given in terms of the MN/m^2 . This is the preferred multiple of the basic unit. Values expressed in this way are, however, numerically equal to values expressed in terms of the N/mm^2 , which is often chosen for convenience of use (i.e. $1 \text{ MN/m}^2 = 1 \text{ N/mm}^2$).

METHODS

1. SCOPE

Clause 2 of this Part of the standard specifies the method for determining the compressive strength of test cubes.

Clause 3 specifies the method for drilling, examination, measurement and testing for compressive strength of cylindrical cores drilled from concrete.

Clause 4 specifies the method for obtaining samples from hardened un-reinforced concrete slabs on site and their preparation as test beams.

Clause 5 specifies the method for determining the flexural strength of moulded or sawn test beams under third point loading.

Clause 6 specifies the method for determining compressive strength, using portions of beams broken in flexure as the test specimens.

Clause 7 specifies the procedure for determining the indirect tensile strength of moulded test cylinders by the splitting method.

2. TEST FOR COMPRESSIVE STRENGTH OF TEST CUBES

NOTE. For the methods of making and curing the cubes see Part 3 of this standard, 'Methods of making and curing test specimens'.

2.1 Apparatus

2.1.1 Compression testing machine. The testing machine may be of any reliable type of suitable capacity for the test and capable of applying the load at the rate specified in 2.2.3. It shall comply as regards repeatability and accuracy with the requirements of BS 1610*.

The testing machine shall be equipped with two steel bearing platens which shall be at least as large as the nominal size of the specimen to which the load is applied.

The upper platen (assuming that the load is applied in a vertical direction) shall be fitted with a ball seating in the form of a portion of a hemisphere, the diameter of which shall not be less than the size of the specimen and not more than three times the size of the specimen. The centre of the sphere shall be within 1.6 mm of the axis of the loading system and within 25 mm of the bearing face of the upper platen. The movable ~~portion shall be held against the~~ spherical seat at all times but the design shall be such that the bearing face can be rotated freely and tilted through small angles in any direction when not under load. The spherical surfaces of the ball seating may be coated thinly with petroleum jelly to prevent corrosion but shall not be lubricated in such a way as to enable movement to occur under load.

NOTE. Lubricants designed to maintain a continuous film under high bearing pressures should not be used in ball seatings of testing machines. The intention is that the seating should enable the platen to accommodate itself initially to the shape of the test specimen but should thereafter be restrained from moving.

The lower compression platen shall be a plain non-tilting bearing block.

Five determinations of surface hardness shall be made, uniformly distributed over the area of the bearing platens to be in contact with the specimen. The average of these values shall not fall below 600 Vickers Hardness Value† and

* BS 1610, 'Methods for the load verification of testing machines'.

† BS 427, 'Method for Vickers hardness test', Part 1, 'Testing of metals'. Where the Vickers testing machine is not available Scale C of the Rockwell hardness test (see BS 891, 'Method for Rockwell hardness test', Part 1, 'Testing of metals') may be used as an alternative, but in the event of a dispute the result of the Vickers test should be the criterion. (See BS 860, 'Tables for comparison of hardness scales'.) The purchaser should note the need to accept the indentations on the faces of new platens resulting from the hardness test.

the range between the highest and the lowest value shall not exceed 50 HV. The flatness tolerance* ('flatness 1') for the area to be in contact with the specimen shall be 0.032 mm wide and the surface texture† shall be 3.2 µm CLA‡.

2.1.2 Auxiliary platens

2.1.2.1 Material. Where the bearing platens of the machine are larger than the specimen, loose steel auxiliary platens may be interposed between the machine platens and the surfaces of the cube to be in contact with them. These platens shall not be less than 12.0 mm or more than 25.0 mm thick. Five determinations of surface hardness shall be made, uniformly distributed over the area of both faces of the auxiliary platens. The average of these values shall not fall below 600 Vickers Hardness Value* and the range between the highest and the lowest value shall not exceed 50 HV.

2.1.2.2 Tolerances. The platens shall be such that they are accurate within the following limits:

(1) *Dimensions.* The distance between either pair of opposite edges, based on the average of four symmetrically placed measurements, shall be the nominal size of the specimen ± 0.125 mm§.

(2) *Flatness.* The flatness tolerance* ('flatness 1') for each face shall be 0.03 mm wide.

(3) *Squareness.* The squareness tolerance* ('squareness 1') for each edge with respect to the adjacent edge as datum edge shall be 0.5 mm wide.

(4) *Parallelism.* The parallelism tolerance* ('parallelism 1') or one face with respect to the other face as datum face shall be 0.06 mm wide.

(5) *Surface texture.* The surface texture† of each face shall be 3.2 µm CLA‡.

2.2 Procedure

2.2.1 General. Cubes stored in water shall be tested immediately on removal from the water, whilst they are still wet. Cubes for testing at 24 h shall be tested in the moist condition and no-fines cubes (see the method of making and curing no-fines test cubes in Part 3 of this standard) shall be tested in the air-dry condition. Surface water and grit shall be wiped off the cube and any projecting fins removed.

2.2.2 Placing the cube in the testing machine. The bearing surfaces of the testing machine and of any auxiliary platens shall be wiped clean and any loose grit or other material removed from the surfaces of the cube which are to be in contact

* See the clause on geometrical tolerances in BS 308, 'Engineering drawing practice'.

† BS 1134, 'Centre-line-average height method for the assessment of surface texture'.

§ This is understood to mean that work with any CLA value from zero to the stated value is acceptable.

§ This tolerance should be considered in conjunction with that for squareness.

with the compression platens. The test cube shall be placed in the machine in such a manner that the load shall be applied to opposite sides of the cube as cast, i.e. not to the top and bottom. The axis of the cube shall be carefully aligned with the centre of thrust of the spherically seated platen. As the latter is brought to bear on the specimen the movable portion shall be rotated gently by hand so that uniform seating is obtained. No packing other than auxiliary steel platens shall be used between the faces of the cube and the platens of the testing machine.

2.2.3 Loading. The load shall be applied without shock and increased continuously at a rate of approximately $15 \text{ MN/m}^2 \text{ min}$ until no greater load can be sustained. The maximum load applied to the cube shall be recorded. The appearance of the concrete and any unusual features in the type of failure shall be noted.

2.3 Calculation. The compressive strength of each cube shall be calculated by dividing the maximum load by the nominal cross-sectional area and shall be expressed to the nearest 0.5 MN/m^2 .

2.4 Report. The following information shall be included in the report on each cube:

- (1) date of test,
- (2) identification mark and nominal size of specimen
- (3) age of cube,
- (4) compressive strength,
- (5) curing conditions, and
- (6) appearance of concrete and type of fracture, if these are unusual.



3. PREPARATION AND COMPRESSION TESTING OF DRILLED CORES FROM CONCRETE

3.1 Preparation

3.1.1 Size of core. The standard size of core shall be 150 mm diameter. If the quality of the concrete is such as to permit drilling without damage, cores with a diameter of 100 mm may be used. The length of the core as received shall be at least 95 % of the diameter.

NOTE. Existing core drills of 152.4 mm and 101.6 mm respectively, may be used provided they are clearly marked.

3.1.2 Drilling. Cores shall be drilled perpendicularly to the surface and in such a manner as not to damage them. Each core shall be clearly marked with its identification immediately after cutting and the orientation of the core shall be recorded.

3.1.3 Examination. Each core shall be examined for the following features and, when required, shall also be photographed.

(1) *Description of aggregate.* Pieces of coarse aggregate shall be examined for group classification, petrological name, particle shape and surface texture according to the details in BS 812*. The maximum size shall be estimated to the nearest appropriate size specified in BS 882†.

(2) *Distribution of materials.* Each core shall be examined for evidence of segregation of the individual materials by visually comparing the approximate coarse aggregate/mortar ratio at different planes in the core.

(3) *Compaction.* Each core shall be examined for the presence of voids, and the position at which any honeycombing begins shall be noted. In describing the compaction of the concrete in a core the following terms shall be used:

Small void. A void measuring not less than 0.5 mm and not more than 3 mm across in any direction.

Medium void. A void having a dimension greater than 3 mm but not greater than 6 mm.

Large void. A void having a dimension greater than 6 mm.

Honeycombing. Interconnected voids arising from, for example, inadequate compaction or lack of mortar.

A core showing voids but no honeycombing is illustrated in Fig. 1 and a core showing both voids and honeycombing in Fig. 2.

In referring to the extent to which voids occur, the number should be related to the area of the cut surface of the core and may be described as 'negligible', 'few', 'considerable' or 'numerous' in accordance with the values shown in Table 1.

TABLE 1. CLASSIFICATION OF THE EXTENT OF VOIDS

	Number of voids per 100 000 mm ² of cut face		
	Small voids	Medium voids	Large voids
Negligible	Less than 40	Less than 4	
Few	40-150	4-15	Less than 2
Considerable	150-400	15-75	2-15
Numerous	More than 400	More than 75	More than 15

3.1.4 *Measurement.* Each core shall be measured for:

(1) *Average diameter.* The average diameter shall be determined from three pairs of measurements, the two measurements in each pair being taken at right

* BS 812, 'Methods for the sampling and testing of mineral aggregates, sands and fillers'.

† BS 882, 'Coarse and fine aggregates from natural sources'.

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angles to each other. One pair shall be at the middle of the core and the other two pairs at the quarter points of the length. The average of these six readings to the nearest 2 mm shall be taken as the average diameter.

(2) *Length.* The maximum and minimum lengths as received and the length after capping (see 3.1.5) shall be determined to the nearest 5 mm.

(3) *Reinforcement.* The position of any reinforcement shall be determined by measuring to the nearest 2 mm from the centre of the exposed bars to the top of the core. The size of the bars and if possible the spacing shall be determined.

3.1.5 End preparation for compression test. If necessary, the core shall be trimmed so that it has ends that are reasonably even, perpendicular to the axis and of the same diameter as the body of the specimen. Both ends of the core shall be capped in accordance with the requirements for the preparation of the surface of hardened concrete in the clause describing the making and curing of test cylinders in Part 3 of this standard. The length after capping shall be not less than the diameter and not greater than twice the diameter.

3.2 Compression testing

3.2.1 Apparatus. The testing machine shall comply in all respects with the requirements of 2.1.

3.2.2 General procedure. The core shall be tested immediately on removal from the water at the expiration of a period of immersion of at least 48 h and whilst it is still wet.

3.2.3 Placing the core in the testing machine. The bearing surfaces of the testing machine and of any auxiliary platens shall be wiped clean and any loose sand or other material removed from the ends of the core. The core shall be placed in the testing machine and its axis carefully aligned with the centre of thrust of the spherically seated platen. As the latter is brought to bear on the core the movable portion shall be rotated gently by hand, so that uniform seating is obtained. No packing other than auxiliary steel platens shall be used between the ends of the core and the platens of the testing machine.

3.2.4 Loading. The load shall be applied without shock and increased continuously at a rate of approximately 15 MN/m² min until no greater load can be sustained. The maximum load applied to the core shall be recorded. The appearance of the concrete and any unusual features in the type of failure shall be noted.

3.3 Calculation. The measured compressive strength of each core shall be calculated by dividing the maximum load by the cross-sectional area, calculated from the average diameter (see 3.1.4(1)). A correction factor according to the length/diameter ratio of the specimen after capping shall be obtained from Fig. 3. The product of this correction factor and the measured compressive

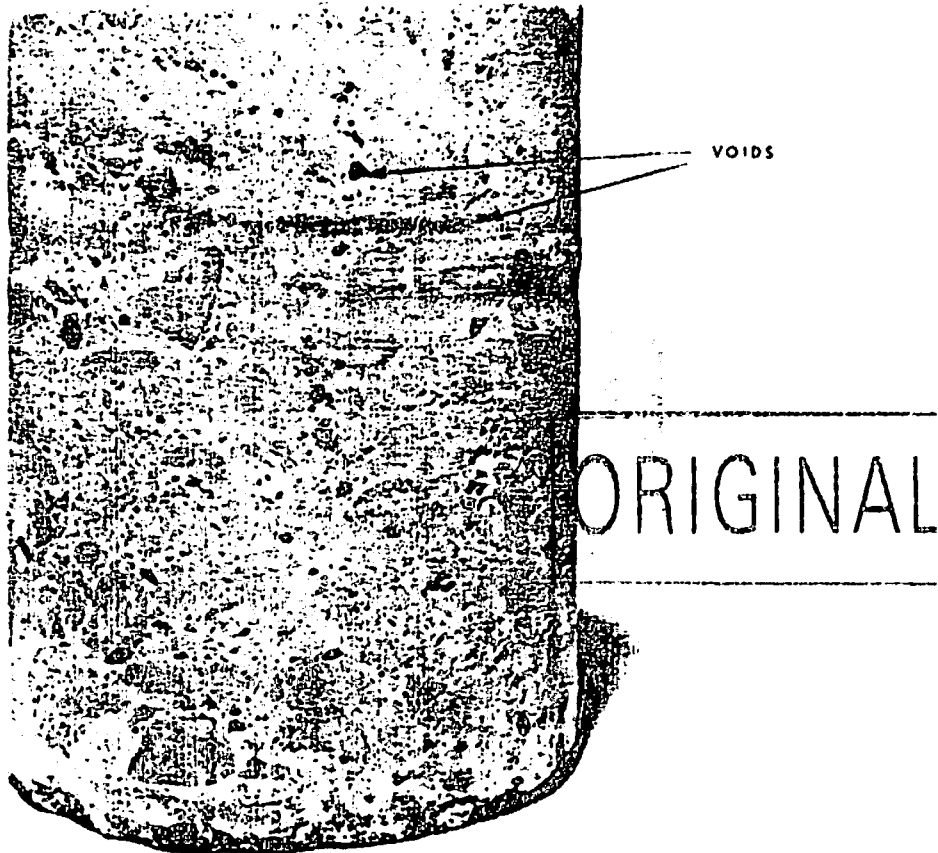


Fig. 1. 150 mm diameter concrete core, showing voids but no honeycombing

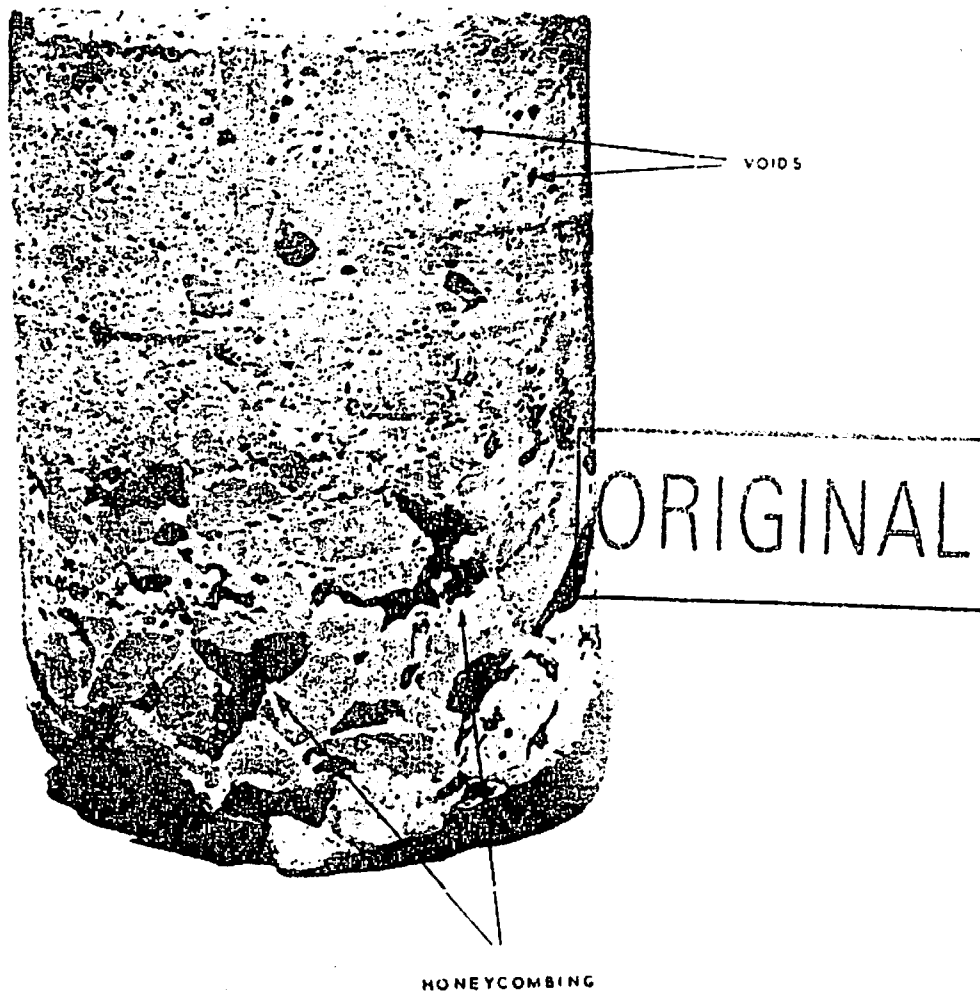


Fig. 2. 150 mm diameter concrete core, showing voids in the upper portion of the core and honeycombing at the bottom

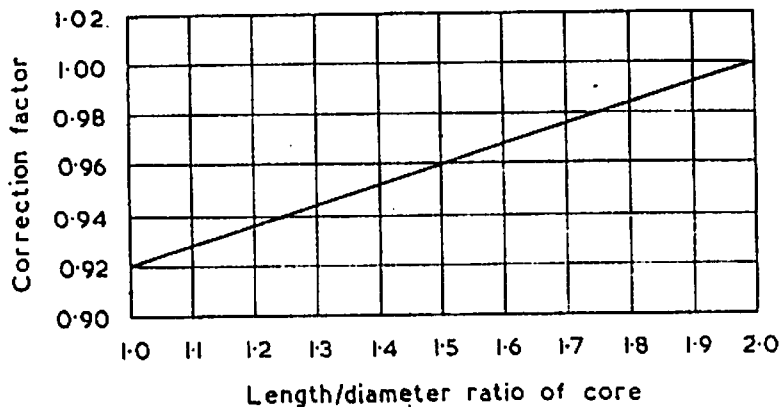


Fig. 3. Correction factor for compression test on a core

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length shall be known as the corrected cylinder strength, this being the equivalent strength of a cylinder having a length/diameter ratio of 2. The cube strength of the concrete shall be estimated by multiplying the corrected cylinder strength by 1.25.

Each of the above results shall be expressed to the nearest 0.5 MN/m².

E. It should be noted that the position of reinforcement in cores cut from reinforced concrete may affect the result.

Report. The following information shall be included in the report on each :

- (1) date of test,
- (2) identification mark,
- (3) age of core, when known, at date of test,
- (4) average diameter,
- (5) maximum and minimum lengths, as received,
- (6) length after capping,
- (7) method of capping,
- (8) measured compressive strength,
- (9) corrected cylinder strength,
- (10) estimated cube strength,
- (11) appearance of concrete and type of fracture, if these are unusual,
- (12) size, position and spacing of any reinforcement,
- (13) description of aggregate, including maximum size, group classification, mineralogical name, particle shape and surface texture,
- (14) distribution of materials, and
- (15) compaction of concrete.

The following supplementary information shall be included in the report, if required:

- (16) copy of the photograph, or photographs, of the core as received.

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4. PREPARATION OF SAWN BEAMS

NOTE. For the methods of testing the specimens see Clause 5. See also the clause describing test for the dynamic modulus of elasticity by an electrodynamic method in Part 5 of this standard.

Size of test beam. The standard size of test specimen shall be 150 mm × 150 mm × 750 mm long. Specimens with cross-sectional dimensions of either more or less than 150 mm × 150 mm down to a lower limit of 100 mm × 100 mm may be used, provided the cross section is square, the distance between the axes of the outer pair of rollers is 4 times the depth of the beam and the overall length is 5 times the depth of the beam.

Samples. Samples shall be obtained in such a way that they are not affected by the means of cutting employed on the site.

4.2.1 Preparation. The sample shall be placed on a soft timber packing to take up base irregularities and the whole shall be firmly wedged and clamped to the bed of a concrete sawing machine so that the original finished surface of the concrete is uppermost and parallel to the machine bed. Two parallel longitudinal saw cuts at right angles to the original finished surface shall be made to the full depth of the sample and 150 mm apart. The sawn portion of the sample shall then be turned on one side and a further saw cut made at a distance of 150 mm from the original finished surface of the concrete. The ends of the sample shall then be trimmed to the required length by sawing at right angles to the other sawn surfaces.

Where the beam is to be tested in flexure in such a manner that the original finished surface of the concrete is in tension (see 5.2.2.2) this surface shall be ground flat at the positions where the rollers of the testing machine make contact with the specimen.

On completion of these operations the beam shall be marked for later identification and submerged in water at a temperature of $20 \pm 1^\circ\text{C}$ for at least 48 h before testing in accordance with the requirements of Clause 5, or of the clause describing the test for the dynamic modulus of elasticity by an electrodynamic method in Part 5 of this standard, as appropriate.

4.2.2 Tolerances. The prepared specimen shall be checked to ensure that it is accurate within the following limits:

(1) *Dimensions.* The depth and width of the beam, each based on the average of six symmetrically placed measurements, shall be the nominal size $\pm 2.0 \text{ mm}^*$ and shall be reported to the nearest 0.5 mm.

(2) *Flatness.* The flatness tolerance† ('flatness 1') for the bearing surfaces at the four positions where the rollers of the testing machine make contact with the specimen shall be 0.25 mm wide.

(3) *Squareness.* The squareness tolerance† ('squareness 1') for each side face with respect to the original finished surface of the concrete as datum face shall be 2.0 mm wide.

(4) *Parallelism.* The parallelism tolerance† ('parallelism 1') for the bottom face with respect to the original finished surface of the concrete as datum face, and for one side face with respect to the other side face as datum face shall be 4.0 mm wide.

5. TEST FOR FLEXURAL STRENGTH

NOTE. For the methods of making and curing moulded beams see Part 3 of this standard. For the method of preparing beams sawn from hardened concrete see Clause 4.

5.1 Apparatus. The testing machine may be of any reliable type of suitable capacity for the test and capable of applying the load at the rate specified in

* This tolerance should be considered in conjunction with those for flatness, squareness and parallelism.

† See the clause on geometrical tolerances in BS 308, 'Engineering drawing practice'.

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5.2.3. It shall comply as regards repeatability and accuracy with the requirements of BS 1610*.

The testing machine shall be equipped with two pairs of steel rollers having a nominal diameter of 38–40 mm and a length at least 10 mm greater than the width of the beam. One pair of rollers shall be used to support the beam and the other to apply the load. All four rollers shall be positioned so that their axes are normal to the axis of the beam under test. The distance between the axes of one pair of rollers shall be 600 mm for 150 mm beams or 400 mm for 100 mm beams. The other pair of rollers shall be located at the third points, i.e. the distance between their axes shall be 200 ± 0.5 mm or 133.3 ± 0.5 mm, respectively. The rollers shall be machined so that the entire cylindrical surface lies inside an annular zone (cylindricity tolerance†) 0.1 mm wide. All rollers shall be mounted in such a manner that the load is applied symmetrically and normal to the axis of the specimen and without subjecting the specimen to any torsional stresses or restraints.

One suitable arrangement which complies with these requirements is shown in Figs. 4 and 5.

5.2 Procedure

5.2.1 *General.* Beams stored in water shall be tested immediately on removal from the water whilst they are still wet. Beams for testing at 24 h shall be tested in the moist condition. If the beams to be tested have already been used for testing the dynamic modulus of elasticity (see Clause 4 of Part 5 of this standard), they shall not remain out of water for more than $\frac{1}{4}$ h and shall be returned to water storage for at least $\frac{1}{4}$ h before testing. If not tested on the same day they shall not remain out of water for more than one hour. Surface water and grit shall be wiped off the beam and any projecting fins removed.

5.2.2 *Placing the beam in the testing machine.* The bearing surfaces of the supporting and loading rollers shall be wiped clean and any loose grit or other material removed from the surfaces of the specimen where they are to make contact with the rollers.

5.2.2.1 *Moulded specimen.* The specimen shall be placed in the testing machine in such a manner that the load shall be applied to a side adjacent to the uppermost surface as cast in the mould, along two lines spaced 200 mm or 133 mm apart. The axis of the specimen shall be carefully aligned with the axis of the loading device. No packing shall be used between the bearing surfaces of the specimen and the rollers.

* BS 1610, 'Methods for the load verification of testing machines'.

† See the figure showing tolerancing errors of cylindricity in BS 308, 'Engineering drawing practice'.